

# Lunar Lander Vibe Coding Project

ChatGPT Vibe Coding History — Lab 4

Student: Abdul Mateen Shaikh

Repository: mat-7697/AM332\_lunarLander

## Task 1: Fix touchdown validation

Mateen — Fix Task 1 in my Lunar Lander project. Safe touchdown must require the ship to be over a landing pad, vertical speed within MAX\_SPEED\_Y, horizontal speed within MAX\_SPEED\_X, and angle within MAX\_ANGLE. Inspect touchdown() and fix the missing validation.

ChatGPT — The existing touchdown() condition checks the landing pad, vertical speed, and angle, but does not check horizontal velocity. The missing condition is `abs(self.vel.x) <= MAX_SPEED_X`.

Corrected condition:  
`if pad and abs(self.vel.x) <= MAX_SPEED_X and abs(self.vel.y) <= MAX_SPEED_Y and abs(angle) <= MAX_ANGLE:`

Commit: Fix touchdown horizontal speed validation  
SHA:  
cc8d8a5784ab22b6f4592e97f03222b5e868e9ed

## Task 1: Verification

Mateen — How should I demonstrate the original bug and fixed behavior?

ChatGPT — Use a case where the ship is over the pad, vertical speed is safe, angle is safe, but horizontal speed is above the limit. Example:  $V_x = 30$ ,  $V_y = 25$ , Angle = 5 degrees. Since  $MAX\_SPEED\_X$  is 25, the landing should now be rejected. Before the fix it could incorrectly succeed because  $V_x$  was not checked.

## Task 2: Implement ship\_color(fuel\_ratio)

Mateen — Implement ship\_color(fuel\_ratio). It receives a value from 0 to 1 representing remaining fuel. Return an RGB tuple and make the ship gradually shift from the normal light color at full fuel toward red at low fuel.

ChatGPT — Implemented a clamped interpolation between full-fuel and empty-fuel colors:

```
def ship_color(fuel_ratio):  
    fuel_ratio = max(0.0, min(1.0, fuel_ratio))  
    full = (230, 230, 240)  
    empty = (255, 70, 70)  
    return tuple(  
        int(empty[i] + (full[i] - empty[i]) * fuel_ratio)  
        for i in range(3)  
    )
```

Commit: Implement fuel-based ship color<br>SHA: 16a8caf27340d91e088d7ca1e5daed6b2de1a6f7

### Task 3: Implement on\_landing(score)

Mateen — Add a visible landing effect through on\_landing(score). Use fireworks, with a stronger effect for higher-value landings, while keeping the existing scoring and landing behavior.

ChatGPT — Implemented a landing fireworks particle effect. A successful landing creates particles around the landing position. Normal landings create 12 particles, while score  $\geq 500$  creates 18 particles.

Commit: Add landing fireworks effect  
SHA: 63f875701b40232324129e46445fd5ec6d162c0b

## Task 4: Implement `bonus_life_threshold()`

Mateen — Implement `bonus_life_threshold()`. It takes no arguments and returns an integer score threshold. The existing `update()` method already handles the bonus-life bookkeeping.

ChatGPT — The surrounding bookkeeping already handles the award, so the hook only needs a fixed threshold. Implemented 1500 points.

```
def bonus_life_threshold():  
    return 1500
```

Commit: Implement bonus life threshold  
SHA: d0a9566ad0786386fa3c14812a3d63e10b2494d2

## Final Testing and Recording

Mateen — What should I test and record now?

ChatGPT — Test all four tasks. Task 1: show  $V_x > 25$  being rejected. Task 2: use thrust and observe the ship becoming redder as fuel decreases. Task 3: perform a valid landing and show fireworks. Task 4: run: `python -c "import game; print(game.bonus_life_threshold())"` and verify 1500. For the after video, prioritize visible features such as fuel color and successful landing/fireworks.

## GitHub Commit Record

Personal repository: [https://github.com/mat-7697/AM332\\_lunarLander](https://github.com/mat-7697/AM332_lunarLander)

1. Fix touchdown horizontal speed validation

cc8d8a5784ab22b6f4592e97f03222b5e868e9ed

2. Implement fuel-based ship color

16a8caf27340d91e088d7ca1e5daed6b2de1a6f7

3. Add landing fireworks effect

63f875701b40232324129e46445fd5ec6d162c0b

4. Implement bonus life threshold

d0a9566ad0786386fa3c14812a3d63e10b2494d2

## Lab 4 Submission Checklist

### COMPLETED

- ✓ Task 1 — touchdown horizontal speed validation
- ✓ Task 2 — fuel-based ship color
- ✓ Task 3 — landing fireworks
- ✓ Task 4 — bonus-life threshold
- ✓ Four separate commits
- ✓ Personal GitHub repository

### SUBMISSION MATERIALS

- Before-changes video — approximately 10 seconds
- After-changes video — approximately 10 seconds
- Complete ChatGPT Vibe Coding chat history exported as PDF
- Updated code
- Lab-4 folder containing the required deliverables
- Final submission

Do not create a pull request to the SETAPESU26 main repository.

End of Chat History — Lunar Lander — Abdul Mateen Shaikh